

# CR-BOT V4

## Hierarchical Policy Architecture for Clash Royale

### Architecture, simulator-first training, and formal evaluation protocol

A technical design proposal for the next CR-BOT policy generation.

**Revision focus.** V4 is a *simulator-policy* project. Raw expert-video behavior cloning is not a required stage because the current extractor can create causally inconsistent trajectories. Equally important, **sim-to-real fidelity is explicitly deferred**: it is not a V4 objective, training gate, or promotion criterion. V4 succeeds or fails by whether its frozen policies become measurably stronger inside a fixed, versioned simulator under a predeclared champion-challenger evaluation protocol.

|                     |                                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Project             | CR-BOT                                                                            |
| Document            | V4 Technical Design Paper                                                         |
| Date                | 8 September 2026                                                                  |
| Status              | Proposed architecture; design targets and budgets require experimental validation |
| Repository baseline | Keschler/cr-bot, reviewed at commit fe69c7da (7 September 2026)                   |

## Abstract

CR-BOT V4 is a proposed redesign of the current recurrent Clash Royale policy. It preserves the strongest properties of V3 - a public-only actor, structured legal-action masks, hierarchical WAIT/PLAY → card → placement decisions, recurrent memory, a privileged training-only critic, deterministic evaluation, and simulator exploit auditing - while changing the information flow and the way tactical competence is learned.

The central architectural changes are: (1) a public state-estimation interface between public game events and the actor; (2) an explicit current-state path to every action head; (3) four hand cards represented as card-conditioned queries over battlefield context; (4) a mandatory full-resolution spatial placement decoder; and (5) learned, interruptible WAIT duration. The central training change is equally important: because the current visual extractor can miss certain spells and produce other incorrect detections, raw expert-video behavior cloning is not trusted as the default bootstrap. V4 instead uses simulator-generated supervised distillation, counterfactual search, search-guided DAgger, PPO curriculum learning, and eventually historical self-play. Expert data remains useful for future perception work and later confidence-gated partial imitation, but it is outside the core V4 training and strength-evaluation loop.

The design goal is not a much larger network. It is a cleaner causal interface, denser and more trustworthy tactical supervision, more faithful temporal actions, and an evaluation system that can distinguish genuine policy improvement from benchmark noise, lucky checkpoints, reward hacking, and regressions.

### Core thesis

CR-BOT should remain a learned neural policy. Rules and simulator search should be used as **teachers, priors, opponents, diagnostics, or evidence sources** - not as hidden strategy inside the deployed actor. Expert replays are deferred to future transfer work. V4 should learn basic tactics from trustworthy synthetic counterfactuals before PPO is asked to solve long-horizon strategy. Every training stage must produce a frozen challenger that can be accepted or rejected against the current champion without changing the benchmark after seeing the result.

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## 1 Baseline and motivation

V4 should be treated as a clean policy generation rather than another compatibility-preserving patch to the V3 checkpoint family. At the repository revision reviewed for this paper, the actor consumes `PolicyObservationV2`; combines a  $32 \times 18$  public raster, public entity tokens, and global features; uses a compact Transformer plus a 256-unit GRU; and decodes a masked hierarchical action: WAIT/PLAY, then card slot, then card-conditioned placement. The critic may consume exact simulator state during training while the actor remains public-only [7].

The current project has already produced useful diagnostic evidence. The repository reports a provisional actor with roughly 96k distillation decisions and a paired held-out result of 16/24 versus 2/24 for an earlier retained baseline, while retaining exploit gates. More importantly, recent diagnostics found concrete architecture-level failure modes: placement behavior could regress despite small global PPO KL, and stale recurrent cheap-cycle preferences could dominate current card information until the explicit public card path became primary [7]. These are signals that representation and action decoding matter at least as much as optimizer tuning.

### 1.1 What V3 already gets right

| Design choice                                                        | Why it matters                                                                                                    | V4 decision     |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------|
| Hierarchical action factorization                                    | Card and placement are conditional, avoiding the independent card/x/y failure mode discussed in other CR RL work. | <b>Retain</b>   |
| Public-only actor inputs                                             | Live deployability is preserved; hidden simulator information cannot become a crutch.                             | <b>Retain</b>   |
| Privileged critic                                                    | Exact training-state information can improve value estimation without entering actor inference.                   | <b>Retain</b>   |
| Recurrent memory                                                     | Necessary for partial observability, opponent cycle, and temporal intent.                                         | <b>Retain</b>   |
| Structured legality masks                                            | Impossible actions are removed without hard-coding strategic choices.                                             | <b>Retain</b>   |
| Deterministic evaluation and exploit audit                           | Prevents reward bugs or simulator loopholes from masquerading as policy strength.                                 | <b>Retain</b>   |
| Multiple direct action-head bypasses accumulated during V3 debugging | Useful diagnostically, but increasingly complex and checkpoint-driven.                                            | <b>Redesign</b> |
| Fixed-tick WAIT                                                      | Creates repeated null decisions and stretches temporal credit assignment.                                         | <b>Redesign</b> |

### 1.2 What the wider Clash Royale AI landscape suggests

The most useful external lesson is not that one public architecture should be copied. It is that strong systems converge on several recurring ideas: explicit state estimation, hierarchical decisions, spatially faithful placement, dense tactical supervision before sparse-reward RL, reproducible simulator infrastructure, and population-based self-play only after a competent policy exists [8, 5, 6, 1].

Table 1: Selected external lessons relevant to V4. Public project claims are treated as engineering evidence, not independent benchmarks.

| Project / line                | Observed strength                                                                                            | Lesson for CR-BOT                                                                                                          |
|-------------------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| KataCR                        | Visual-only mobile agent using detection/OCR and offline sequence/RL methods.                                | Visual deployment is feasible; perception and policy must share a latency-aware contract.                                  |
| ClashAI                       | Shared engine/live observation ideas, expert imitation, spatial tokens, search-teacher and DAGger direction. | Tactical search can provide denser placement/card supervision than pure sparse-reward PPO.                                 |
| Jason-XII simulator           | Fast Gymnasium-style simulator with PPO experimentation.                                                     | Action representation can become the bottleneck; hierarchical conditional decisions are preferable to independent outputs. |
| High-throughput CR simulators | Large-scale deterministic simulation infrastructure.                                                         | Throughput is useful only when paired with sealed benchmarks, reproducible seeds, and exploit-safe evaluation.             |
| AlphaStar / league RL         | Supervised bootstrap followed by historical-policy and exploiter populations.                                | Population self-play is a later-stage robustness tool, not the bootstrap.                                                  |

### 1.3 The expert-video constraint changes the training design

#### Known CR-BOT constraint: current expert replays are not automatically BC-safe

The current visual extractor does not reliably recover every action/state transition: certain spells can be missed, and other detections can be wrong or noisy. This is more dangerous than ordinary independent label noise. A missed spell can alter unit HP, deaths, positions, elixir, and all subsequent state-action pairs, creating a trajectory that is causally impossible.

Consider a real sequence in which an expert casts Fireball, kills support troops, then deploys Hog Rider. If Fireball is missed, the extracted trajectory can show support troops disappearing without a corresponding action and then label the subsequent Hog play from a state that never existed. Training harder on that example teaches the policy an extractor artifact.

Therefore V4 adopts the following rule:

#### Data trust rule

**Raw expert-video behavior cloning is disabled for V4.** The V4 policy is trained and evaluated from simulator-grounded data only. Expert footage remains a resource for the later transfer/perception project; it does not enter V4 policy loss or V4 promotion evidence.

## 2 V4 design objectives

1. **Preserve the public-information boundary.** The actor uses only information that is public in the game plus deterministic public rules; exact simulator state remains critic-only.
2. **Make current state first-class.** Every action head receives immediate battlefield features directly, not

only through recurrent memory.

3. **Preserve exact board geometry for placement.** Placement is decoded from a board-aligned feature map.
4. **Treat each hand card as a conditional tactical query.** Card selection evaluates each available card against the current threat/opportunity context.
5. **Represent time explicitly.** WAIT predicts a duration and can be interrupted by important public events.
6. **Bootstrap from trustworthy supervision.** Simulator teacher targets and counterfactual search replace unverified expert BC as the default first policy-training stage.
7. **Learn from the student’s own failures.** DAgger queries the teacher on states actually visited by the current policy.
8. **Make improvement falsifiable.** Champion–challenger comparisons, paired cells, sealed promotion suites, and confidence bounds determine whether a checkpoint actually improved.
9. **Scale after correctness.** Larger models and very large self-play budgets are deferred until controlled ablations justify them.

### 3 End-to-end V4 architecture

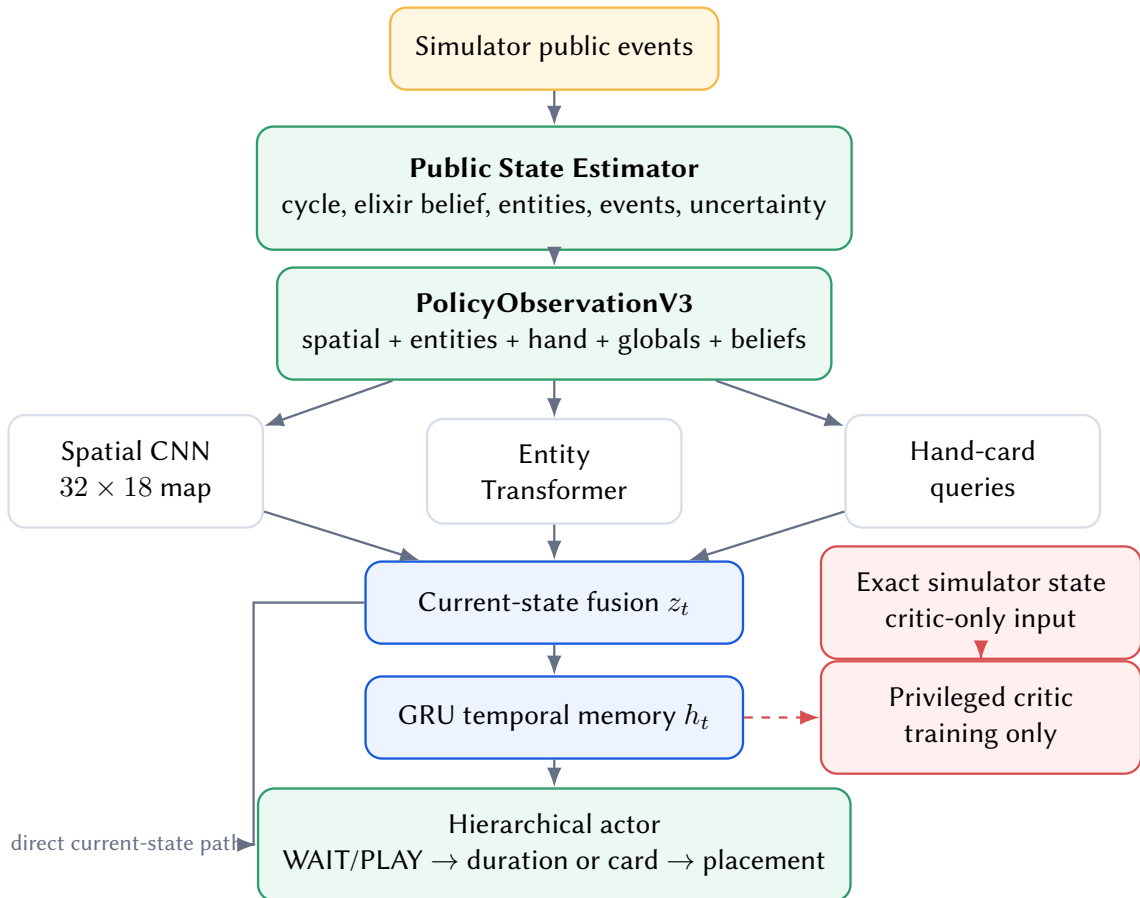


Figure 1: Core V4 information flow inside the simulator. The actor sees only public state; exact hidden simulator state is restricted to the training-only critic. Live perception and sim-to-real transfer are intentionally outside the V4 evaluation scope.

### 3.1 Public State Estimator

For V4, the estimator is a deterministic/probabilistic state machine between **simulator public events** and the policy. It performs public bookkeeping that should not have to be rediscovered by the GRU, such as observed card cycle, elixir bounds, entity age, and event history. The actor never receives hidden simulator state. A future live adapter may later target the same `PolicyObservationV3` contract, but proving that mapping is accurate is not part of V4.

Recommended estimator outputs include:

- visible unit identity, side, position, velocity, age, HP, status, and current target when public;
- public tower HP and activation state;
- exact visible own hand, next card, elixir, and public legality facts;
- observed opponent cards, possible hand/cycle set, and cards definitely out of cycle;
- opponent elixir interval/distribution updated from observed spending and public regeneration;
- regulation/overtime and elixir-rate phase;
- recent deployment, death, damage, target-change, and action events;
- explicit uncertainty only for genuinely hidden-but-inferable public beliefs, especially opponent hand/cycle and elixir.

The estimator should estimate *state*, not strategy. It must not output “play Cannon now” or any equivalent action recommendation. Deterministic game bookkeeping such as cycle arithmetic is appropriate; strategic decision rules are not.

### 3.2 PolicyObservationV3

`PolicyObservationV3` should be versioned and immutable, produced by one shared builder used by simulator training, search, DAgger, PPO, and frozen evaluation. A future live bridge can target the same schema without changing the V4 policy contract.

Table 2: Recommended V3 observation components. Dimensions are starting points, not fixed requirements.

| Component             | Recommended shape                       | Purpose                                                                                                      |
|-----------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Spatial tensor        | 24–32 channels<br>$\times 32 \times 18$ | Occupancy, HP/threat fields, tower zones, motion/age, and public legality channels.                          |
| Entity tokens         | up to $128 \times (32-64)$              | Visible units/towers with type, coordinates, HP, velocity, age, side, status, and public target information. |
| Hand tokens           | 4 tokens, 64–128 latent                 | Card identity, cost, slot, affordability, availability, and card-class features.                             |
| Global scalars        | compressed to 128–256 latent            | Time, own elixir, tower summaries, match phase, masks, and recent action statistics.                         |
| Opponent belief       | structured distributions                | Possible hand/cycle and elixir belief derived from public evidence only.                                     |
| Event/history summary | short ring buffer or encoded summary    | Deployments, damage, deaths, target changes, and action timing.                                              |

### 3.3 Spatial encoder

V4 should always preserve a board-aligned feature map. A small residual CNN is sufficient: for example  $24\text{--}32$  input channels  $\rightarrow 64 \rightarrow 64/96$  channels while keeping the  $32 \times 18$  grid. A pooled summary can feed global fusion, but the unpooled map must remain available to placement decoding.

This is a deliberate departure from relying on global pooling for the action path. Kiting, defensive building pull geometry, spell value, troop spacing, and bridge placements are local geometric problems; their representation should remain local until the cell score is produced.

### 3.4 Entity and hand encoding

Visible entities pass through a compact Transformer, initially around model dimension 128, four heads, and two layers. This retains the useful V3 representation for a variable set of public units.

The four hand cards are encoded separately. Each hand token should contain card identity embedding, elixir cost, slot position, affordability, and optional stable card-class properties. The four tokens are not merely indices: they become candidate action queries.

### 3.5 Current-state fusion and recurrent memory

Let  $z_t$  denote the fused current public state and  $h_t$  the recurrent memory:

$$z_t = \text{Fuse}(S_t, E_t, H_t, G_t, B_t), \quad (1)$$

$$h_t = \text{GRU}(z_t, h_{t-1}). \quad (2)$$

Every action head receives both  $z_t$  and  $h_t$ . This makes V4 explicitly residual in time: immediate card, elixir, threat, and geometry information cannot be lost merely because the GRU carries a stale historical preference.

### 3.6 Hierarchical action decoder

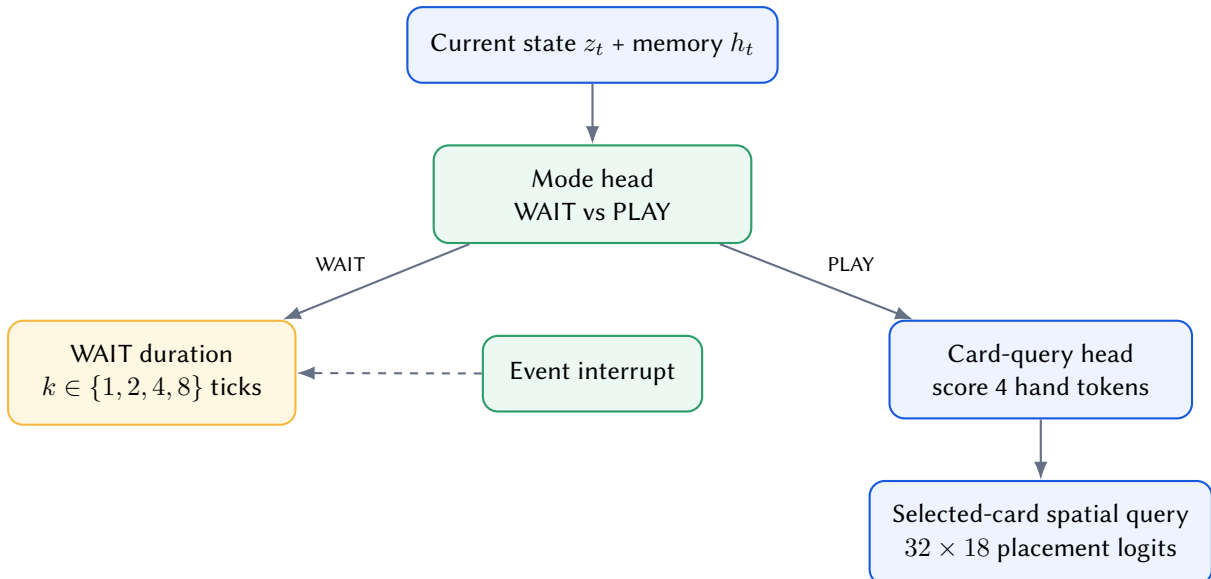


Figure 2: V4 action factorization. Legality masks remove only impossible actions; timing, card choice, and placement remain learned.

#### 3.6.1 Mode head

The mode head predicts whether a card should be committed now. It receives  $z_t$ ,  $h_t$ , and a compact public legality summary. Affordability need not be rediscovered from memory when it is already known.



### 3.6.2 Card-query head

For PLAY, each of the four hand-card tokens becomes a query over battlefield/entity context. This makes card choice relational rather than a four-logit readout from one undifferentiated state vector. A Cannon token can attend to an approaching win condition; Fireball can attend to clustered targets; Hog can attend to building state and pressure context. The network learns those relations rather than having them hard-coded.

### 3.6.3 Card-conditioned spatial placement

After a card is selected, its query conditions the  $32 \times 18$  spatial map. A simple and efficient decoder is a projected dot product:

$$q_a = f_q(e_{\text{card}}, z_t, h_t), \quad (3)$$

$$\ell_{x,y} = \langle W_q q_a, W_k S_t[x, y] \rangle + b_{x,y}, \quad (4)$$

$$P(x, y \mid a, \text{PLAY}) = \text{MaskedSoftmax}(\ell, M_a). \quad (5)$$

This preserves exact placement geometry while computing only one selected-card map.

## 3.7 Learned, interruptible WAIT duration

A fixed high-frequency WAIT action creates many nearly identical transitions. V4 instead treats WAIT as a semi-Markov action with a small duration head, initially  $k \in \{1, 2, 4, 8\}$  base decision ticks. A long wait ends early if a significant public event occurs, for example an enemy deployment, tower damage, a newly visible threat, a substantial tracker change, or a legality/affordability transition that should trigger reconsideration.

#### Why this matters

Duration prediction reduces the overwhelming frequency of null actions, improves the semantic meaning of one policy decision, reduces inference/rollout waste, and lets reward discounting depend on elapsed game time rather than the number of polling frames.

## 3.8 Privileged critic and belief auxiliaries

The privileged critic remains training-only. It may consume exact simulator state while actor inputs stay public-only, following asymmetric actor-critic principles [4]. Actor features should be detached from the privileged branch when required to make information separation explicit.

Low-weight auxiliary heads may predict opponent elixir buckets, possible hand membership, or next-card probabilities from public history. Their purpose is representation learning for partial observability, not action supervision.

### 3.9 Recommended initial capacity

| Block               | Starting configuration           | Reason                                                                |
|---------------------|----------------------------------|-----------------------------------------------------------------------|
| Spatial map         | 64–96 channels at $32 \times 18$ | Preserve local geometry without making placement expensive.           |
| Entity Transformer  | $d = 128$ , 4 heads, 2 layers    | Similar scale to V3; increase only after ablation.                    |
| Hand/card embedding | 64–128                           | Only four candidate queries; enough for relational scoring.           |
| Fused current state | 256                              | Shared direct path into all action heads and GRU.                     |
| GRU                 | 256, 1 layer                     | Keep V3 recurrent scale initially.                                    |
| Placement decoder   | projected per-cell compatibility | Full resolution, card-conditioned, efficient.                         |
| Target actor size   | approximately 4–8M parameters    | Architecture and supervision are the hypothesis, not parameter count. |

## 4 Training methodology

The V4 training path is deliberately simulator-first. The policy is improved in stages, but **no stage is trusted because its training loss improved**. Each stage must survive the same frozen evaluation machinery before it can replace the current champion.

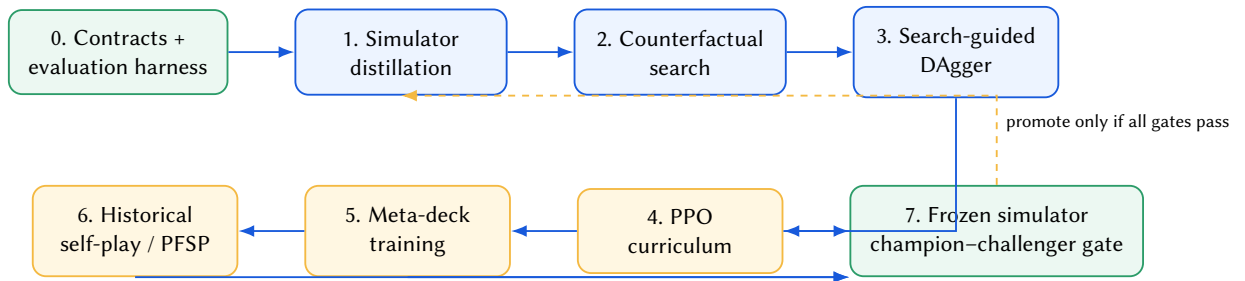


Figure 3: Core V4 training loop. Sim-to-real fidelity, live shadow play, and expert-video imitation are deferred beyond V4 and do not affect V4 promotion.

### 4.1 Stage 0 - freeze contracts and build the evaluation harness

Before serious V4 training:

- freeze the simulator/ruleset revision, PolicyObservationV3, action semantics, legality-mask semantics, and contract hashes for each experiment generation;
- freeze the incumbent **champion** checkpoint and its exact evaluation manifest;
- build three distinct evidence sets: a reusable development suite, a permanent important-state regression bank, and a sealed promotion-suite generator;
- define the primary strength metric, practical improvement margin, no-regression tolerances, confidence level, and sample-size rule *before* evaluating a challenger;
- ensure every evaluation cell is reproducible from a manifest containing opponent, deck, side, starting hand/state, simulator seed, policy seed, and horizon;

- verify that training reward, PPO losses, KL, entropy, teacher agreement, or Elo cannot by themselves promote a checkpoint.

#### Scope boundary

V4 does **not** answer whether simulator skill transfers to the live game. No real-vs-simulator comparator, perception-noise calibration, degraded-observation test, or live ladder result is required for V4 promotion. Those become a separate later transfer project built on a frozen V4 champion. Keeping the questions separate prevents an unresolved deployment problem from blocking policy research and prevents live noise from making simulator progress impossible to measure cleanly.

### 4.2 Stage 1 - simulator teacher distillation

V4 should first learn basic tactical behavior from simulator-generated states rather than unverified expert video. The state generator should use the existing short-state curriculum and broaden it deliberately across offense, ground defense, air defense, spell value, kiting, cycling, low-elixir situations, counterpushes, and bridge defense.

For each state, a teacher provides a supervised target. Early in development that teacher can be a mixture of validated tactical rules and simulator consequences; over time, counterfactual search should become the authority.

#### Directional starting budget

A practical initial target is roughly **0.5–2 million supervised decision states**, generated from many short, randomized tactical scenarios. This is a planning range, not a claimed optimum. Promotion depends on held-out tactical performance, not a fixed sample count.

Supervised targets include mode, card, selected-card placement, and WAIT duration. Soft placement targets may spread probability across cells with near-equivalent simulated consequences.

### 4.3 Stage 2 - counterfactual search teacher

The simulator offers something a replay cannot: counterfactual actions from the exact same state. At a student-visited state  $s$ , build a candidate set from the student's top proposals, selected tactical alternatives, and WAIT-duration choices. Clone  $s$ , apply each candidate, simulate a short horizon, and score the resulting state.

$$\mathcal{A}(s) = \text{TopK}_{\pi}(s) \cup \mathcal{A}_{\text{explore}} \cup \mathcal{A}_{\text{WAIT}}, \quad (6)$$

$$Q_H(s, a) = U(s, a, s_H) + \lambda V_{\text{ref}}(s_H), \quad (7)$$

$$\pi_T(a \mid s) = \text{softmax}(Q_H(s, a)/\tau). \quad (8)$$

$U$  should be based on signed consequences such as own/enemy tower damage, removed threat, surviving unit value, elixir swing, and immediate tactical danger.  $V_{\text{ref}}$  supplies continuation value when the short horizon cannot resolve long-term consequences.

Candidate count and horizon are compute controls. A sensible first search is top 8–16 student actions plus a small number of structured alternatives, with 2–5 seconds of simulated continuation. Difficult states can receive deeper or multi-horizon evaluation.

#### 4.4 Stage 3 - search-guided DAgger

Behavior cloning from a fixed dataset suffers from state-distribution shift. DAgger instead collects states induced by the current student and asks a stronger teacher what should be done there [2]. This is especially valuable for CR-BOT because the policy creates unusual failure states that expert players rarely visit.

One round is:

1. Roll out the current actor in short scenarios and full matches.
2. Select states with high uncertainty, high consequence, teacher disagreement, or observed tactical failure.
3. Generate candidate actions from the current actor plus exploration alternatives.
4. Branch those candidates in the simulator and produce a soft teacher distribution.
5. Store state, teacher distribution, horizon, consequence metrics, generator/seed identity, and source metadata.
6. Distill a new student from the aggregated dataset.
7. Re-evaluate on frozen states and actor-controlled matches.

This directly turns the actor’s own mistakes into supervised corrections.

#### 4.5 Stage 4 - PPO strategic fine-tuning

PPO should begin only after V4 can already execute basic tactics. PPO is then used for long-horizon choices that short search cannot reliably teach: pressure timing, when to hold elixir, safe investment, accepting damage, opposite-lane pressure, cycle management, spell cycling, matchup adaptation, and overtime risk [3].

Table 3: Directional PPO/self-play scale after the supervised/search bootstrap.

| Phase                | Directional budget                     | Purpose                                                                                           |
|----------------------|----------------------------------------|---------------------------------------------------------------------------------------------------|
| Mechanics foundation | 1–5M decisions                         | Short tactical states; heavy distillation/Dagger rehearsal; offense, defense, spells, kite/cycle. |
| Scripted curriculum  | 10–30M additional                      | Passive, win-condition, reactive, aggressive, and randomized opponents.                           |
| Meta-deck training   | 30–100M additional                     | Representative archetypes → broader validated meta pool; weakness-prioritized sampling.           |
| Historical self-play | 100–300M additional                    | PFSP over frozen historical policies plus scripted anchors and exploiters.                        |
| Small league         | 300M–1B+ cumulative, only if justified | Main learner, exploiters, 16–32 historical policies, payoff matrix, periodic resets.              |

#### 4.6 Stage 5 - reference protection during PPO

Small global KL is not sufficient evidence that behavior is preserved. V4 should track policy movement per action factor and on a curated important-state buffer:

- separate KL/entropy for mode, WAIT duration, card choice, and selected-card placement;
- frozen defense, spell, cycle, low-elixir, and overtime states;
- branch divergent candidate actions to measure actual consequences;

- optional state-weighted reference-policy KL as a regularizer, not the sole safety gate;
- quarantine if decisive state behavior regresses even when average reward rises.

#### 4.7 Stage 6 - historical self-play and PFSP

Only after the policy is strong against scripted/meta anchors should training move toward a population. A small historical pool sampled with prioritized fictitious self-play (PFSP) reduces overfitting to the newest opponent and exposes non-transitive weaknesses. Large AlphaStar-style league machinery is unnecessary until simpler populations demonstrably saturate [1].

#### 4.8 Stage 7 - frozen simulator promotion

At any meaningful training milestone, the candidate checkpoint is frozen and becomes a *challenger*. It is not promoted because reward increased or because it beat one opponent. It enters the formal protocol in Section 7. If any mandatory gate fails, the incumbent champion remains unchanged. Once a promotion suite has been revealed, that suite is considered *spent*: it may become diagnostic data, but it may never again serve as unseen promotion evidence for a later challenger.

### 5 Optimization objectives

#### 5.1 Supervised simulator-distillation loss

For a teacher-labeled sample:

$$L_{\text{sup}} = w_m \text{CE}(m) + w_w \text{CE}(k \mid \text{WAIT}) + w_c \text{CE}(c \mid \text{PLAY}) \quad (9)$$

$$+ w_p \text{CE}(x, y \mid c, \text{PLAY}) + w_b L_{\text{belief}} + w_v L_{\text{value}}. \quad (10)$$

Sampling and weighting should be factor-aware so rare decisive PLAY/card/placement targets are not drowned by WAIT states. Natural-distribution evaluation must remain separate from balanced training batches.

#### 5.2 Search distillation

When the teacher provides a soft action distribution:

$$L_{\text{search}} = D_{\text{KL}}(\pi_T(\cdot \mid s) \parallel \pi_\theta(\cdot \mid s)). \quad (11)$$

Soft targets are useful when several placements are tactically equivalent or nearly equivalent.

#### 5.3 PPO with elapsed-time discounting

PPO uses the clipped surrogate objective [3]:

$$r_t(\theta) = \frac{\pi_\theta(a_t \mid o_t)}{\pi_{\text{old}}(a_t \mid o_t)}, \quad (12)$$

$$L_{\text{PPO}} = \mathbb{E}[\min(r_t A_t, \text{clip}(r_t, 1 - \epsilon, 1 + \epsilon) A_t)]. \quad (13)$$

For variable WAIT durations, discount by elapsed game time:

$$\gamma_t = \gamma_{\text{base}}^{\Delta t / \Delta t_{\text{base}}}. \quad (14)$$

GAE should use the corresponding variable discount. This avoids treating an eight-tick wait as if it represented the same temporal transition as a one-tick decision.

## 5.4 Reference and entropy regularization

A practical total objective may include:

$$L_{\text{total}} = -L_{\text{PPO}} + \beta_{\text{ref}} D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) - \beta_H H(\pi_{\theta}) + L_{\text{aux}}. \quad (15)$$

The important-state consequence regression suite remains the promotion authority; averaged KL is only a diagnostic/regularizer.

## 6 Why this architecture and training method should work better

### 6.1 Immediate facts no longer compete with stale memory

V3 diagnostics show that a recurrent preference can dominate current card information. Supplying both  $z_t$  and  $h_t$  to every head makes temporal memory additive rather than authoritative.

### 6.2 The action representation matches the causal decision

Clash Royale decisions are naturally conditional: act now or not; if acting, choose the card; then choose a location compatible with that card. V4 mirrors this structure and strengthens it with card queries and spatial scoring.

### 6.3 Placement learns from geometry rather than global reconstruction

Precise card placement is local. Keeping a full-resolution map until cell scoring reduces the information bottleneck between battlefield geometry and action location.

### 6.4 WAIT becomes a temporal decision rather than a dominant class

Duration prediction compresses repeated null decisions and aligns credit assignment with elapsed game time. Event interruption preserves responsiveness.

### 6.5 Simulator supervision is causally clean

Within the frozen V4 simulator contract, the simulator provides a state, action, and resulting state that are causally and mechanically self-consistent. V4 deliberately evaluates only within that contract. Whether the contract matches the live game is a different question for later. By contrast, a missed event in extracted expert footage can silently break the causal chain even before any policy learning begins.

### 6.6 Counterfactual search is richer than one observed expert action

A replay reveals only what the expert did. State cloning can compare “Cannon one cell left”, “Skeletons”, “WAIT 0.4 seconds”, or “push opposite lane” from the exact same state. That produces a dense preference signal and can discover alternatives that are as good as or better than the observed action.

### 6.7 DAgger trains on the policy’s actual failure distribution

An expert rarely reaches the bizarre states caused by a weak bot. DAgger does, because it labels the student’s own states. This is especially valuable during early V4 deployment in the simulator.

### 6.8 PPO is reserved for long-horizon strategy

PPO is no longer asked to discover every exact defensive cell from delayed tower damage. It receives a tactically competent policy and can focus its gradient budget on strategy under the student’s state distribution.

## 7 Formal evaluation: deciding whether V4 actually improved

### The central evaluation rule

A V4 checkpoint is “better” only if a **frozen challenger** beats the current **frozen champion** on predeclared, paired, simulator-only evidence while satisfying no-regression and integrity guardrails. Training reward, teacher agreement, PPO KL, entropy, loss curves, Elo, or a hand-picked set of wins are diagnostics; none is a strength claim by itself.

This distinction is essential. A policy can increase training return by exploiting a reward bug, improve mean win rate by becoming much better against easy opponents while collapsing against one archetype, or look strong because the same 24 cells have been inspected and tuned against repeatedly. V4 therefore treats evaluation as a versioned experimental protocol rather than a final command run after training.

### 7.1 Two different questions must not be conflated

**Checkpoint question:** is challenger  $C$  better than incumbent champion  $B$ ? This is answered by one sealed champion–challenger promotion experiment.

**Method question:** is the V4 architecture/training recipe better than V3 or an ablated V4 recipe? This requires several independent training seeds under equal budgets and compares the *distribution of learning outcomes*, not the single best checkpoint.

A checkpoint can be promoted without proving the whole V4 method is superior. Conversely, an architecture claim requires evidence across seeds even if one individual run is excellent.

### 7.2 Evaluation cell and paired comparison

Define one evaluation cell

$$x_i = (\text{scenario/opponent, deck, side, initial state/hand, environment seed, policy seed, horizon}). \quad (16)$$

The challenger and champion are run on the **same** cell  $x_i$ . For a complete match, define score

$$g(\pi, x_i) = \begin{cases} 1 & \text{win,} \\ \frac{1}{2} & \text{draw,} \\ 0 & \text{loss.} \end{cases} \quad (17)$$

The paired improvement is

$$d_i = g(C, x_i) - g(B, x_i). \quad (18)$$

Using the same cell for both policies removes large amounts of variance from starting hand, opponent behavior, scenario randomization, and environment stochasticity. Side-sensitive tests are duplicated with the side assignment reversed. If deployment uses deterministic action selection, the primary evaluation uses deterministic action selection as well; stochastic-policy evaluation is reported separately.

### 7.3 The four benchmark layers

| Layer             | Contents                                                    | Role                                                                  |
|-------------------|-------------------------------------------------------------|-----------------------------------------------------------------------|
| Development suite | Small visible tactical states and matches, freely reusable. | Debugging and fast iteration only. Never cited as promotion evidence. |



| Layer                           | Contents                                                                                                                          | Role                                                                                                                              |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Important-state regression bank | Previously discovered catastrophic states plus representative offense, defense, spell, timing, cycle, and overtime states.        | Permanent guardrail. It may be inspected repeatedly, so it tests preservation, not unseen generalization.                         |
| Sealed tactical promotion set   | Newly generated held-out states stratified by tactical family. Candidate and champion actions are branched from identical states. | Measures local decision quality and detects rare decisive regressions. The exact cells are hidden until the challenger is frozen. |
| Sealed full-match promotion set | Paired complete matches against frozen scripted, meta-deck, historical-policy, and adversarial anchor strata.                     | Primary strategic strength evidence. A used set is retired from future promotion.                                                 |

#### Why sealed suites matter

If the same “held-out” matrix is run after every change, it gradually becomes training information even if no gradient is computed on it. V4 therefore distinguishes *reusable validation* from *unseen promotion evidence*. Once promotion results are inspected, those exact cells are spent.

A practical implementation does not need a permanently secret file. Freeze the benchmark *generator* and its stratum weights first; after the challenger checkpoint hash is fixed, sample a fresh seed manifest, hash that manifest, then run it unchanged. The candidate cannot be selected using those exact cells because they did not exist until after the checkpoint was frozen.

#### Disposition of the current 6-cell and 24-cell matrices

The existing exact and shuffled matrices remain valuable because they contain known behavior and past failure cases, but their results have already been inspected repeatedly. In V4 they should be promoted to the **regression/development bank**, not reused as unseen promotion evidence. Future “better than champion” claims require fresh sealed cells sampled only after the challenger is frozen.

### 7.4 Tactical evaluation should compare consequences, not teacher agreement

For a tactical state  $s_i$ , let  $a_C$  and  $a_B$  be the actions selected by challenger and champion. Clone the same state and evaluate the two first actions with an identical continuation protocol. Define a predeclared consequence score  $F$  from simulator facts relevant to that scenario family, for example prevented own-tower damage, enemy-tower damage, removed threat, surviving unit value, and elixir swing. Then

$$t_i = F(s_i, a_C) - F(s_i, a_B). \quad (19)$$

The continuation controller and horizon are identical for both branches. This isolates whether the candidate’s decision is better rather than merely whether it matches a teacher. A second integrated test lets each policy continue itself; that measures end-to-end tactical execution. Both are useful and answer different questions.

The tactical evaluation score is **not the shaped training reward**. Each scenario family declares its evaluation outcome before training, using directly observable simulator consequences. Where one scalar would require arbitrary trade-offs, report the outcome vector (for example tower HP preserved, residual threat, and elixir difference) and define the primary component lexicographically or by a frozen normalization.



The evaluation definition cannot be edited after a challenger has been trained.

Each tactical family is reported separately. A macro tactical score gives every family equal weight:

$$T_{\text{macro}}(\pi) = \frac{1}{K} \sum_{k=1}^K \frac{1}{N_k} \sum_{i \in k} F(\pi, x_i). \quad (20)$$

This prevents a large number of easy offense states from hiding a collapse in air defense or spell usage.

### 7.5 Full-match primary metric

Let  $K$  be the predeclared opponent/deck strata. The primary match metric is a macro-average of per-stratum match score:

$$S_{\text{macro}}(\pi) = \frac{1}{K} \sum_{k=1}^K \frac{1}{N_k} \sum_{i \in k} g(\pi, x_i), \quad \Delta_{\text{match}} = S_{\text{macro}}(C) - S_{\text{macro}}(B). \quad (21)$$

Macro-averaging is primary because the evaluation distribution should not be dominated by whichever opponent happens to have the most cells. Micro-average win rate, crown differential, normalized Princess-Tower HP differential, average match duration, and per-opponent score are reported as secondary metrics.

Direct challenger-vs-champion matches are also run with side swaps. They are a useful cross-check, but not the only benchmark: mirror head-to-head can favor a specialist that performs poorly against the broader opponent pool.

### 7.6 Catastrophic regression metric

Average improvement is insufficient when a new policy sometimes makes disastrous decisions. For each important/tactical state define a predeclared severe-loss threshold  $\tau$  in consequence units. A catastrophic regression occurs when

$$t_i \leq -\tau. \quad (22)$$

The catastrophic-regression rate is

$$R_{\text{cat}} = \frac{1}{N} \sum_i \mathbf{1}[t_i \leq -\tau]. \quad (23)$$

Thresholds are scenario-family-specific and fixed before evaluation. Examples include losing at least a declared amount of Princess-Tower HP within the branch horizon, converting a defended threat into a crown loss, or turning a safe spell trade into a clearly negative elixir/tower consequence. Action disagreement alone is never classified as regression.

### 7.7 Statistics: confidence bounds, not a single observed percentage

All primary comparisons use paired cells and a predeclared sample size. The report includes a one-sided 95% lower confidence bound (LCB) for the candidate-minus-champion improvement. A stratified paired bootstrap over evaluation blocks is the default because it naturally preserves the benchmark strata and paired structure. One block contains all deliberately linked variants of the same base draw, such as both side swaps; bootstrap the block rather than pretending those games are independent. For family-level no-regression statements, use Holm-corrected one-sided intervals/tests or an equivalent simultaneous procedure.

The promotion sample size is chosen *before* the challenger result is visible. A pilot development run may estimate the standard deviation  $\sigma_d$  of paired differences. For a target detectable improvement  $\delta$  and desired

power  $1 - \beta$ , an initial approximation is

$$N \approx \left( \frac{(z_{1-\alpha} + z_{1-\beta})\sigma_d}{\delta} \right)^2, \quad (24)$$

followed by rounding upward to satisfy minimum counts in every stratum. The pilot estimates variance, not the candidate’s true effect. Optional stopping is not allowed unless a sequential test with its own corrected boundaries is specified in advance.

### 7.8 Stage acceptance is not the same as global champion promotion

During bootstrap, a distillation or DAgger checkpoint may be useful enough to seed the next training stage even if it is not yet the strongest full-match policy. V4 therefore uses two decisions:

- **Stage acceptance:** the checkpoint improves the metric targeted by that stage (for example paired tactical consequence) and passes integrity/regression guardrails. It may seed the next stage.
- **Global champion promotion:** the checkpoint passes the complete full-match champion–challenger rule below. Only this changes the official strongest policy.

This prevents an early tactical learner from being mislabeled as the project champion while still allowing a clean supervised  $\rightarrow$  DAgger  $\rightarrow$  PPO progression.

### 7.9 Formal champion–challenger promotion rule

Before the first V4 promotion attempt, choose practical margins  $\epsilon_{\text{match}}$ ,  $\epsilon_{\text{tact}}$ , per-stratum regression tolerance  $\rho$ , and catastrophic-regression tolerance  $\rho_{\text{cat}}$ . They are policy requirements, not values tuned after looking at a candidate. A sensible first version can use a small positive match margin rather than merely  $> 0$ , then revise thresholds only when the entire benchmark protocol is versioned.

A challenger is promoted only when **all** mandatory gates pass:

| Gate                                 | Formal requirement                                                                                                                                                                                                                        |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| G0 - validity                        | Same sealed simulator/ruleset and observation/action contracts; public-only actor; complete horizons; deterministic replay; zero unexplained rejected/fallback actions; exploit audit clean. Any failure invalidates the strength result. |
| G1 - primary match strength          | $\text{LCB}_{95}(\Delta_{\text{match}}) > \epsilon_{\text{match}}$ . This is the primary promotion endpoint.                                                                                                                              |
| G2 - tactical strength               | $\text{LCB}_{95}(\Delta T_{\text{macro}}) > \epsilon_{\text{tact}}$ for a tactical-stage promotion, or at minimum is non-negative for later strategic stages.                                                                             |
| G3 - no hidden archetype collapse    | For every mandatory opponent/tactical stratum, the simultaneous lower bound on candidate-minus-champion score is greater than $-\rho$ .                                                                                                   |
| G4 - catastrophic regressions        | The upper confidence bound on $R_{\text{cat}}(C) - R_{\text{cat}}(B)$ is below $\rho_{\text{cat}}$ . Critical hand-curated states may additionally be zero-tolerance.                                                                     |
| G5 - direct head-to-head cross-check | Challenger does not lose the paired, side-swapped direct match set; report its confidence interval separately. This is a guardrail, not a substitute for G1.                                                                              |
| G6 - behavior integrity              | No collapse of card usage, WAIT duration, placement diversity, or legal action support that indicates a degenerate policy. These are diagnostics unless a predeclared hard safety bound is violated.                                      |

### Do not promote on training metrics

Higher PPO return, lower loss, higher teacher agreement, lower KL, higher entropy, higher Elo, or more crowns on one script may explain *why* a policy changed. They cannot prove that it is stronger. Elo is especially unsuitable as the primary gate because its value depends on the opponent population and can obscure non-transitive matchups. Promotion is based on frozen outcome/consequence benchmarks only.

## 7.10 Champion lifecycle and benchmark retirement

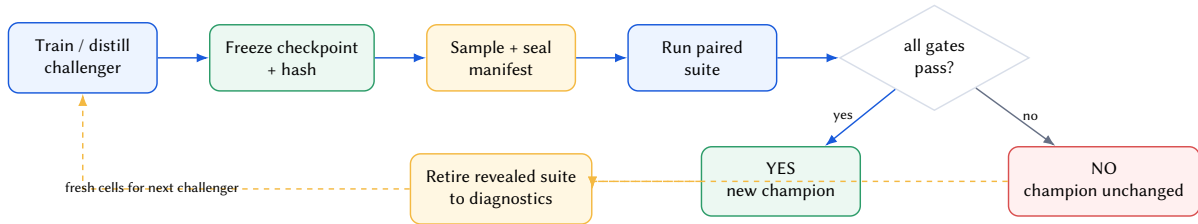


Figure 4: Champion–challenger lifecycle. A revealed promotion suite is retired regardless of pass/fail; the next challenger receives fresh sealed cells.

The current champion is immutable during one evaluation generation. When a challenger passes, its checkpoint hash becomes the new champion baseline. V3 remains a permanent historical anchor even after the first V4 promotion. This creates a monotonic chain of evidence: every accepted checkpoint has defeated the immediately preceding champion under the protocol that was current at the time.

## 7.11 Evaluating whether the V4 method is better

A single promoted V4 checkpoint does not prove that the architecture or training recipe is better. To make that claim, compare complete training recipes under equal conditions:

- same simulator/ruleset, scenario generator, opponent schedule, initial checkpoint class, and total environment-decision budget;
- same evaluation milestones, for example after fixed cumulative decision counts rather than wall-clock time;
- at least three independent training seeds for exploratory architecture decisions; use more if variance remains large;
- no cherry-picking the best seed or best checkpoint after the fact.

Two endpoints matter. First, final frozen strength at budget  $D$ . Second, sample efficiency across the whole run. Save checkpoints at predeclared decision milestones  $d_j$ . After *all* compared training runs are complete and frozen, sample a fresh method-evaluation manifest and evaluate every saved milestone on those same cells. This avoids using the final method test to steer training while still allowing a leakage-free learning curve. If  $S_j$  is the resulting score, define normalized area under the learning curve

$$\text{AULC} = \frac{1}{D} \sum_{j=1}^{m-1} \frac{S_j + S_{j+1}}{2} (d_{j+1} - d_j). \quad (25)$$

A V4 method claim should report paired seed differences in final strength, AULC, and seed-to-seed variance. Training recipes should also receive the same hyperparameter/tuning budget; otherwise extra tuning can be mistaken for an architectural gain. A design that occasionally produces one excellent checkpoint but

frequently collapses is not clearly better than a slightly weaker but reproducible method.

### 7.12 Ablation protocol

Ablations answer which V4 ideas actually matter. Run them under the same training budget and benchmark generation:

- full V4 vs no direct current-state path;
- card-query head vs simple four-logit MLP card head;
- full-resolution spatial placement vs pooled/MLP placement;
- fixed WAIT tick vs learned WAIT duration;
- simulator distillation only vs distillation + search vs search-guided DAgger;
- PPO from distillation checkpoint vs PPO from DAgger checkpoint;
- privileged critic vs public-only critic;
- latest-opponent mirror self-play vs fixed historical/PFSP pool.

For architectural claims, report final score and AULC over matched seeds. For training-stage claims, also report the amount of environment experience and search compute required to obtain the gain. This prevents a component from appearing “better” only because it received more supervision.

### 7.13 What every evaluation report must contain

Every promotion artifact should be machine-readable and include:

- champion/challenger checkpoint hashes, code revision, simulator/ruleset hash, observation/action contract hash;
- predeclared benchmark-manifest hash, sample-size rule, confidence level, practical margins, and all stratum definitions;
- one row per paired cell with seeds, side, opponent, deck, outcome, crowns, tower HP, rejected/fallback actions, and terminal reason;
- tactical branch consequence records and catastrophic-regression classifications;
- macro and per-stratum estimates, paired confidence intervals, head-to-head results, and gate pass/fail reasons;
- exploit-audit result and deterministic replay hashes for any failure or surprising divergence;
- an explicit statement that the checkpoint was frozen before the sealed promotion cells were revealed.

This report, rather than an informal “17–7” or “16/24” summary, becomes the evidence used to decide whether CR-BOT improved.

## 8 Implementation plan

V3 should remain frozen as a benchmark and opponent while V4 is developed in parallel. Checkpoint compatibility should not dictate V4 internals; interoperability should exist at the observation, action, and environment boundaries.

| Step | Workstream                  | Deliverable                                                                                                                        | Gate                                                              |
|------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| 1    | Observation contract        | PolicyObservationV3, simulator-public event schema, uncertainty fields, deterministic contract hash.                               | No model change yet.                                              |
| 2    | State estimator             | Simulator-public cycle/elixir belief, event memory, and actor input construction.                                                  | Unit tests and deterministic reset/replay traces.                 |
| 3    | Evaluation harness          | Champion registry, development suite, important-state bank, sealed promotion-suite generator, paired statistics, promotion report. | Protocol frozen before V4 strength claims.                        |
| 4    | V4 actor                    | Spatial map, entity Transformer, hand queries, current-state fusion, GRU, hierarchical heads.                                      | Supervised only initially.                                        |
| 5    | Simulator distillation      | State generator, factorized targets/losses, matched-seed evaluation.                                                               | Pass stage-specific tactical gate.                                |
| 6    | Search teacher              | State cloning, candidate branching, short-horizon consequence score, soft labels.                                                  | Teacher improves paired held-out consequences.                    |
| 7    | Dagger                      | Student-state collection, selective query policy, aggregated dataset.                                                              | Student inherits teacher gain without important-state regression. |
| 8    | PPO V4                      | Variable-time GAE, per-head diagnostics, important-state guardrails, curriculum.                                                   | Frozen challenger passes simulator promotion.                     |
| 9    | Meta + historical self-play | Frozen policy pool, directional payoff matrix, PFSP.                                                                               | Only after fixed anchor strength.                                 |
| 10   | Method ablations            | Matched-budget V3/V4 and component experiments across independent training seeds.                                                  | Final strength + AULC + variance reported.                        |
| 11   | Future transfer project     | Live estimator calibration, sim-to-real fidelity, extractor reliability, live evaluation, verified expert ingest.                  | Explicitly outside V4.                                            |

## 8.1 Suggested module boundaries

```

simulator/
  observation_v3.py
  public_state_estimator.py
  evaluation/
    champion_registry.py
    benchmark_manifest.py
    tactical_suite.py
    full_match_suite.py
    paired_statistics.py
    promotion_report.py
  rl/
    model_v4.py
    losses_v4.py
    simulator_teacher.py
    search_teacher.py
    dagger.py
    ppo_v4.py
    important_state_buffer.py
  league/

```

## 8.2 What not to change initially

- Do not replace the GRU with a large temporal Transformer before testing the new state/action representation.
- Do not aggressively increase actor size. Architecture and supervision are the primary hypotheses.
- Do not broaden the player deck before the fixed 2.6 Hog path is strong and transferable.
- Do not let rules execute learner actions in ordinary PPO; use them as teacher proposals, opponents, or test oracles.
- Do not start 300M+ self-play while timing, placement, tactical strength, or promotion reliability remain unresolved.
- Do not add raw expert BC to V4 because more data is available. Any verified expert imitation belongs to the later transfer project.

## 9 Risks and failure modes

| Risk                                | Failure signature                           | Mitigation                                                            |
|-------------------------------------|---------------------------------------------|-----------------------------------------------------------------------|
| Search teacher is short-term greedy | Over-defense, bad long-term elixir trades.  | Continuation value, multiple horizons, full-match teacher validation. |
| Estimator becomes hidden strategy   | Symbolic layer starts recommending actions. | Strict estimator API: state/belief/events only.                       |
| WAIT duration misses events         | Long waits fail to react to fast threats.   | Event interruption and maximum-duration cap.                          |

| Risk                                           | Failure signature                                        | Mitigation                                                                        |
|------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------|
| Card-query head overfits fixed deck            | Strong 2.6 Hog but poor portability.                     | Generic card identity embedding and architecture; deck specificity stays in data. |
| Spatial decoder reduces throughput             | Per-cell scoring becomes expensive.                      | Shared spatial keys; compute placement only after selected card.                  |
| Belief target leaks hidden information         | Actor learns impossible opponent knowledge.              | Public-history target generation; privileged critic remains separate.             |
| Simulator search exploits bugs                 | Teacher systematically rewards unrealistic actions.      | Search branches use the same exploit and deterministic-validity gates as PPO.     |
| PPO erases tactical skill                      | Rare decisive regressions despite higher average return. | Important-state buffer, per-head KL, branch consequences, rehearsal, low LR.      |
| Observation robustness becomes over-randomized | Policy learns to ignore useful precise state.            | Fit noise to measured distributions; ablate noise strength.                       |
| Expert segment admitted too early              | Impossible post-spell states enter supervised loss.      | Event contamination windows and explicit resynchronization.                       |
| Too many objectives interfere                  | Head-specific gradients fight each other.                | Introduce auxiliaries incrementally; log per-head gradients; ablate.              |

## 10 Expected outcome and success definition

V4 is expected to improve **sample efficiency, tactical precision, strategic strength, diagnosability, and regression resistance**, not necessarily produce an immediate step-function in ladder rank. The strongest expected improvements are:

- more stable card selection when hand order changes;
- better defensive, spell, and kiting placement because spatial information remains explicit;
- fewer redundant WAIT decisions and cleaner temporal credit assignment;
- faster tactical improvement because counterfactual search supplies dense preference targets;
- better recovery from student-created failure states through DAgger;
- fewer policy regressions hidden by low global KL;
- no dependence on unreliable expert extractor trajectories.

### Success criterion

V4 is successful if it becomes **easier to improve and harder to regress inside the simulator**. A checkpoint improvement must survive sealed tactical and full-match champion–challenger tests with confidence bounds. A method improvement must reproduce across training seeds, improve final frozen strength and/or area under the learning curve, and avoid unacceptable archetype or catastrophic-state regressions. Sim-to-real transfer is intentionally not part of this definition.

### 10.1 Why not a final rule graph?

Public rule-based agents show that explicit state estimation and engineered decision structure can produce strong play. That is evidence for a clean perception/state/policy separation, not evidence that CR-BOT should cap itself at the designer’s rule set. V4 borrows explicit bookkeeping and teacher structure while retaining a neural policy that can improve from search, RL, and self-play.

### 10.2 Why not pure PPO?

Pure PPO must simultaneously discover timing, card choice, exact placement, and long-horizon strategy from delayed outcomes. V4 deliberately moves low-level tactical learning into simulator distillation and search-guided DAgger. PPO is then used where its on-policy nature is most valuable: optimizing strategy under the policy’s own state distribution.

### 10.3 Why not raw expert BC?

Because current extraction errors can be causally systematic. A missed spell is not merely a wrong label; it can make the following state sequence impossible. Until the visual extractor and resynchronization pipeline are sufficiently reliable, simulator-grounded counterfactual supervision is more trustworthy for policy bootstrapping.

## 11 Conclusion

CR-BOT V4 should not be a larger V3. It should be a cleaner information and learning architecture. The proposed system preserves the current project’s strongest ideas - public-only recurrent acting, factorized legal actions, a privileged critic, deterministic evaluation, and exploit gates - while reorganizing them around five principles:

1. one public state-estimation interface with explicit uncertainty;
2. direct current-state access for every action head;
3. card-conditioned spatial decisions;
4. learned, interruptible timing;
5. a trustworthy supervised-to-RL progression built from simulator distillation and counterfactual DAgger rather than unverified expert-video BC.

The recommended default training order is therefore:

#### Recommended V4 training order

**Simulator teacher distillation → counterfactual search → search-guided DAgger → PPO curriculum → meta-deck training → historical self-play / PFSP, with frozen champion-challenger evaluation at every meaningful promotion point.**

Sim-to-real transfer and verified expert imitation are **future work beyond V4**. They should start only from a frozen simulator champion, so deployment uncertainty cannot contaminate the question of whether the policy itself improved.

The engineering bet is that the next major gain will come from better representation and better supervision, not simply from a larger model or vastly more PPO steps. If the ablations support that hypothesis, V4 provides a scalable path from strong 2.6 Hog tactical play toward broader meta-level self-play while producing a rigorously measured simulator champion that can later be taken into a separate sim-to-real transfer phase.



## A Proposed V4 training checklist

| Milestone                                | Required evidence                                                                                                                                            |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before V4 data generation                | ObservationV3/action contracts frozen; champion checkpoint/hash frozen; development suite, important-state bank, and sealed promotion generator implemented. |
| Before simulator-distillation promotion  | Challenger frozen before promotion cells are revealed; paired tactical improvement; no action collapse; validity/exploit gate clean.                         |
| Before search-teacher labels are trusted | Search teacher improves paired branch consequences over the frozen student on held-out student states; scoring is exploit-clean.                             |
| Before DAgger promotion                  | Student inherits teacher improvement on unseen tactical states and does not regress on the important-state bank.                                             |
| Before PPO                               | DAgger checkpoint is stronger than the incumbent champion under the declared tactical/full-match gates; variable-time GAE tests pass.                        |
| Before meta-deck expansion               | Full-match macro score improves with no mandatory archetype below the predeclared regression tolerance.                                                      |
| Before historical self-play              | Scripted/meta anchors are strong; historical pool initialized; payoff evaluation is fixed and Elo is not the promotion metric.                               |
| Before any checkpoint promotion          | G0–G6 pass on a sealed paired suite; report includes confidence bounds and manifest hashes; used promotion suite is retired.                                 |
| Before claiming V4 method superiority    | Matched-budget V3/V4 experiments across at least three seeds; report final frozen strength, AULC, seed variance, and required ablations.                     |
| Future transfer work                     | Sim-to-real fidelity, live extractor robustness, shadow/live evaluation, and verified expert imitation begin only after V4 simulator evaluation is complete. |

## B V3 to V4 mapping

| V3                                              | V4                                                      | Reason                                                                                         |
|-------------------------------------------------|---------------------------------------------------------|------------------------------------------------------------------------------------------------|
| PolicyObservationV2                             | PolicyObservationV3                                     | Adds explicit uncertainty, belief, and event fields while preserving the public-only boundary. |
| Raster encoder with pooled path                 | Always-on board-aligned spatial map plus pooled summary | Placement never loses local geometry.                                                          |
| Entity Transformer                              | Entity Transformer retained                             | Existing representation is sound; extend features rather than replace.                         |
| Wide/direct card-head variants                  | Four explicit hand-card query tokens                    | One clean generic card-selection mechanism replaces accumulated bypasses.                      |
| GRU-centric decoding plus optional current skip | Current state + GRU supplied to every head              | Prevents stale-memory dominance by construction.                                               |
| Fixed WAIT tick                                 | WAIT duration head plus event interruption              | Reduces redundant decisions and improves temporal credit.                                      |

| V3                                            | V4                                                                     | Reason                                                                                           |
|-----------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Card-conditioned placement                    | Full-resolution card-conditioned spatial decoder                       | Strengthens existing factorization with exact board features.                                    |
| Privileged critic                             | Privileged critic retained                                             | Useful training asymmetry without deployment leakage.                                            |
| Label-only strategic diagnostics + PPO        | Simulator distillation → search/Dagger → PPO                           | Dense tactical learning occurs before sparse-reward strategy optimization.                       |
| Raw expert BC considered as bootstrap         | Expert data is confidence-gated optional augmentation                  | Current extractor misses some spells and contains other noisy detections.                        |
| Small fixed held-out matrices + exploit audit | Champion–challenger protocol + sealed promotion suites + exploit audit | Makes improvement statistical, paired, reproducible, and resistant to repeated benchmark tuning. |

## C Future transfer work: expert data admission protocol

This appendix is deliberately outside the V4 policy-development and promotion path. Its purpose is to preserve a future route to expert imitation without silently reintroducing the current extraction problem once live transfer is addressed.

### A. Per-factor confidence

A sample stores independent validity masks for mode, card, placement, timing, observed entities, and belief targets. Loss is applied only to factors whose prerequisites are verified.

### B. Contamination windows

Any unresolved event that can alter future state - especially missed/ambiguous spells, tower damage, deaths, or card identity - marks the trajectory as contaminated from that time onward.

### C. Resynchronization

Supervision resumes only after enough independent public evidence re-establishes a consistent state. Possible conditions include reliable hand/cycle reconstruction, known elixir bounds, matched tower HP, reconciled visible entities, and a stable event history window.

### D. Admission levels

| Level | Use                                         | Requirement                                                                                               |
|-------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| E0    | Perception debugging only                   | Raw extracted footage; no policy loss.                                                                    |
| E1    | Distribution statistics / shadow evaluation | State estimates usable in aggregate; action trajectory not necessarily causal.                            |
| E2    | Factor-masked imitation                     | Specific action factors verified; no unresolved causal contamination.                                     |
| E3    | Full sequence imitation                     | High-confidence state/action sequence with validated resynchronization and no unresolved critical events. |

**Default policy**

V4 should be completed and evaluated without expert policy loss. E2/E3 expert supervision belongs to the later transfer project; it is not a missing prerequisite for a successful V4 simulator policy.

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